04 The Overshoot Room

Delayed density dependence: cycles & chaos

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Crowding Room (02).
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Overshoot tab

Many organisms breed once a year, so this year's crowding sets next year's numbers. That one-generation lag is enough to break the tidy approach to K. Nothing random is ever added in this room: every wobble comes from the density dependence itself.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain how a time lag in density dependence causes overshoot.
- Describe the sequence from smooth approach to damped oscillation, cycles and chaos as r rises.
- Contrast scramble (Ricker) with contest (Beverton–Holt) competition.
- Explain sensitivity to initial conditions in a deterministic system.

The extension adds

- Use the slope of the map at K ($1 - r$ for the Ricker) to predict where stability is lost.
- Read a cobweb diagram and a bifurcation diagram.
- Estimate a prediction horizon under chaos.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Lecture on chaos (20 minutes):** steps 2 and 4 projected.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 From glide to wobble

8 min

SET UP

Ricker (scramble), K = 500, N_0 = 30, 80 generations. r = **0.5**, then r = **1.8**.

PREDICT FIRST

Sketch both runs. Which approaches K smoothly, and which overshoots?

RUN

Run both.

LOOK AT

the generation-by-generation dots and the *Long-run behaviour* panel.

EXPLAIN

Why does a faster-growing population overshoot K?

2 Cycles, then chaos

12 min

SET UP

Ricker. r = **2.1**, then **2.6**, then **2.8**.

PREDICT FIRST

For each, will the population settle, cycle, or never repeat?

RUN

Run each, reading the *Long-run behaviour* panel and the gold line on the bifurcation diagram.

EXPLAIN

Describe the sequence you saw. Where on the bifurcation diagram does each run sit?

3 Contest competition

8 min

SET UP

Beverton–Holt (contest), r = 2.8.

PREDICT FIRST

Will this population cycle or become chaotic at the same high rate?

RUN

Run.

EXPLAIN

In contest competition some individuals get enough and others get nothing; in scramble everyone gets a share. Why does that difference decide whether the population can overshoot?

4 Two populations, one thousandth apart

7 min

SET UP

Ricker, $r = 2.8$. **Sensitivity test**: On.

PREDICT FIRST

The second population differs by one thousandth of an individual. Will the two stay together?

RUN

Run. Then repeat at $r = 1.8$.

EXPLAIN

How can a system with no randomness be unpredictable? Why does the same tiny difference not matter at $r = 1.8$?

EXTENSION · MSC

5 Where stability is lost

10 min

SET UP

Ricker; use the History Scrubber to step one generation at a time and watch the cobweb.

PREDICT FIRST

For $N' = Ne^{r(1 - N/K)}$, the slope at $N = K$ is $1 - r$. The equilibrium is stable when $|1 - r| < 1$. At what r does the approach start to oscillate, and at what r does the equilibrium become unstable?

RUN

Check with runs at $r = 0.9$, 1.1, 1.9 and 2.1, and on the bifurcation diagram.

EXPLAIN

Explain, using the cobweb, why a slope between 0 and -1 gives damped oscillation and a slope below -1 gives growing ones.

6 The discrete logistic

10 min

SET UP

Discrete logistic, $r = 1.8$, 2.2, 2.5 and 2.7.

PREDICT FIRST

Will this map follow the same sequence as the Ricker at the same r? What could go wrong at high r that cannot happen with the Ricker?

RUN

Run each, and watch the bifurcation diagram.

EXPLAIN

Compare the shapes of the two maps on the cobweb. Why can the discrete logistic send N below zero?

7 A prediction horizon

10 min

SET UP

Ricker, sensitivity test on. $r = 2.8$, then $r = 3.4$.

PREDICT FIRST

How many generations until the two populations are visibly different, at each r?

RUN

Run and step through with the scrubber to find the generation where they part.

EXPLAIN

If you could measure a real population to within 0.1%, how far ahead could you forecast it under chaos? What does this mean for interpreting irregular census data?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Why might a long-lived animal with overlapping generations be less prone to these cycles than an annual insect?
2. BSc Lemming and vole numbers fluctuate dramatically. Is this room's explanation the only possible one? Name another room that offers an alternative.
3. BSc "Chaotic" in everyday speech means random. Why is that not what it means here?
4. MSc Robert May's 1976 paper showed this behaviour in a simple map. Why was that result so surprising to ecologists at the time?
5. MSc Real census data contain noise. How could you tell deterministic chaos from random fluctuation in a time series?
6. MSc Explain why Beverton–Holt competition is monotone for every r, using the shape of the map.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** At r = 0.5 the population glides into K; at 1.8 it overshoots and oscillates with shrinking swings.
- **Step 2.** About r = 2.1: a two-generation cycle. About 2.6: a four-cycle (it splits again near 2.53). About 2.8: chaos, never repeating.
- **Step 3.** Beverton–Holt approaches K smoothly at any r: contest competition cannot overcompensate, so there is nothing to overshoot.
- **Step 4.** At r = 2.8 the two runs part within a few dozen generations; at 1.8 they stay together, because small differences shrink at a stable equilibrium.
- **Step 5.** Oscillation begins above r = 1 (the slope turns negative) and the equilibrium becomes unstable above r = 2.
- **Step 6.** The discrete logistic follows a similar sequence, a little faster (two-cycle above 2, four-cycle near 2.45, chaos near 2.57), and at high r it can produce negative N, which stops the run.
- **Step 7.** The higher the r, the faster the two diverge. A 0.1% measurement error grows to the size of the population within a few tens of generations under chaos.